

MICHAEL B. LAW

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EDUCATION

Massachusetts Institute of Technology

2022–27 (*expected*)

Ph.D. Mathematics.

Research areas: Geometric analysis and partial differential equations.

Advisor: Prof. William P. Minicozzi II.

Australian National University

2021

Bachelor of Mathematical Sciences (Honors), GPA 7.00/7.00.

Thesis title: *Uniqueness of tangent flows in mean curvature flow*.

Advisor: Prof. Ben Andrews.

University of Melbourne

2020

Bachelor of Science (Mathematics & Statistics), weighted average 94/100.

PREPRINTS AND PUBLICATIONS

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1. *Uniqueness of asymptotically cylindrical steady gradient Ricci solitons*. Preprint. [[arxiv:2505.20576](https://arxiv.org/abs/2505.20576)].
 2. *Drift-harmonic functions with polynomial growth on asymptotically paraboloidal manifolds*. Preprint. [[arxiv:2501.05119](https://arxiv.org/abs/2501.05119)].
 3. *Concavity for elliptic and parabolic equations in locally symmetric spaces with nonnegative curvature* (with Shrey Aryan). *Calc. Var. Partial Differential Equations*. **64**, No. 202 (2025). [[arxiv:2403.16783](https://arxiv.org/abs/2403.16783)].
 4. *Positive mass and Dirac operators on weighted manifolds and smooth metric measure spaces* (with Isaac M. Lopez and Daniel Santiago). *J. Geom. Phys.* **209**, No. 105386 (2025). [[arxiv:2312.15441](https://arxiv.org/abs/2312.15441)].

AWARDS AND HONORS

Simons Dissertation Fellowship in Mathematics, Simons Foundation

2025–27

Croucher Scholarship for Doctoral Study, Croucher Foundation

2023–25

Hartley Rogers Jr. Prize, MIT

2023, 25

Presidential Fellowship, MIT

2022–23

Dixson Prize, University of Melbourne

2020

Maurice H. Belz Prize, University of Melbourne

2019

TEACHING AND MENTORING

As tutorial (recitation) leader:

- Calculus, MIT Sep–Dec 2025
- Discrete Mathematical Models, Australian National University (ANU) Mar–Jun 2021, 22
- Advanced Analysis 1, ANU Feb–May 2022
- Mathematics and Applications 1, ANU Aug–Nov 2021

As research mentor:

- Summer Program in Undergraduate Research, MIT Jun–Aug 2023, 25

- Research Science Institute, MIT
- Undergraduate Research Opportunities Program, MIT

Jul 2024

Jun–Aug 2024

As reading program mentor:

- Directed Reading Program, MIT

Jan 2023, 24, 25

INVITED TALKS

University of Toronto Geometry and Topology Seminar	<i>Nov 2025</i>
Lehigh University Geometry Seminar	<i>Nov 2025</i>
Rutgers University Geometric Analysis Seminar	<i>Oct 2025</i>
Columbia University Geometric Analysis Seminar	<i>Oct 2025</i>
UC Berkeley Differential Geometry Seminar	<i>Sep 2025</i>

CONFERENCES ATTENDED

Rutgers Gauge Theory, Low-Dimensional Topology, and Geometric Analysis Conference	<i>May 2025</i>
Southern California Geometric Analysis Seminar	<i>Feb 2025</i>
Challenges in Analysis and Geometry	<i>Jun 2024</i>
Séminaire de Mathématiques Supérieures	<i>Jun 2024</i>
Croucher Symposium	<i>Dec 2023</i>
Southern California Geometric Analysis Seminar	<i>Mar 2023</i>
Geometric Analysis Seminar for Boston Area Graduate Students	<i>Oct 2022</i>